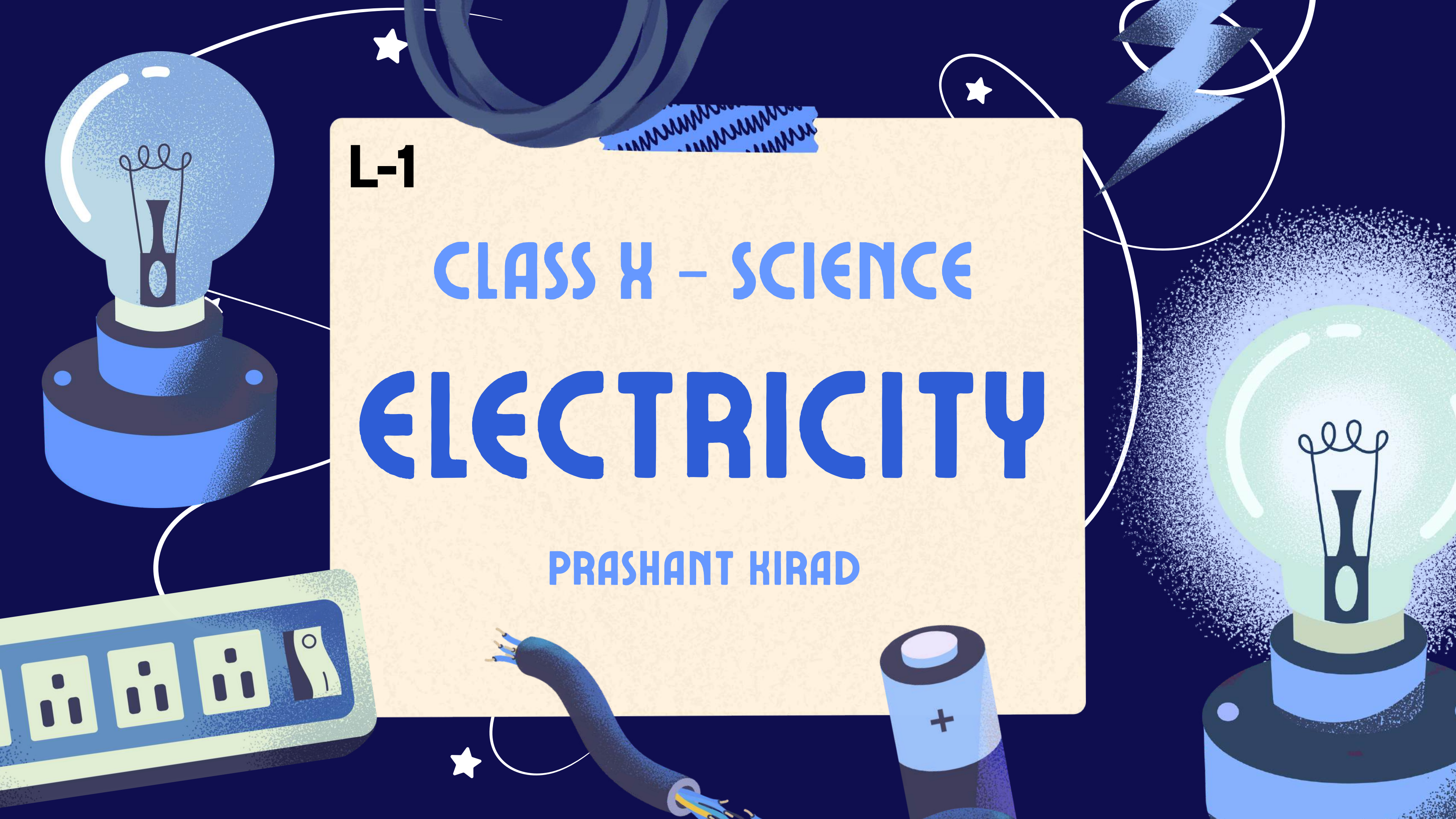


L-1

CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD



JOSH METER



★ CHAPTER FLOW



Current
↓
Potential Difference
↓
Circuit Symbols
↓
Ohm's Law
↓
Resistance & Resistivity
↓
Series/Parallel
↓
Heating Effect
↓
Power

ELECTRICITY

Electricity is a form of energy produced due to the flow or presence of electric charges.



Static electricity

Study of charges at rest

Dynamic electricity

Study of charges in motion.

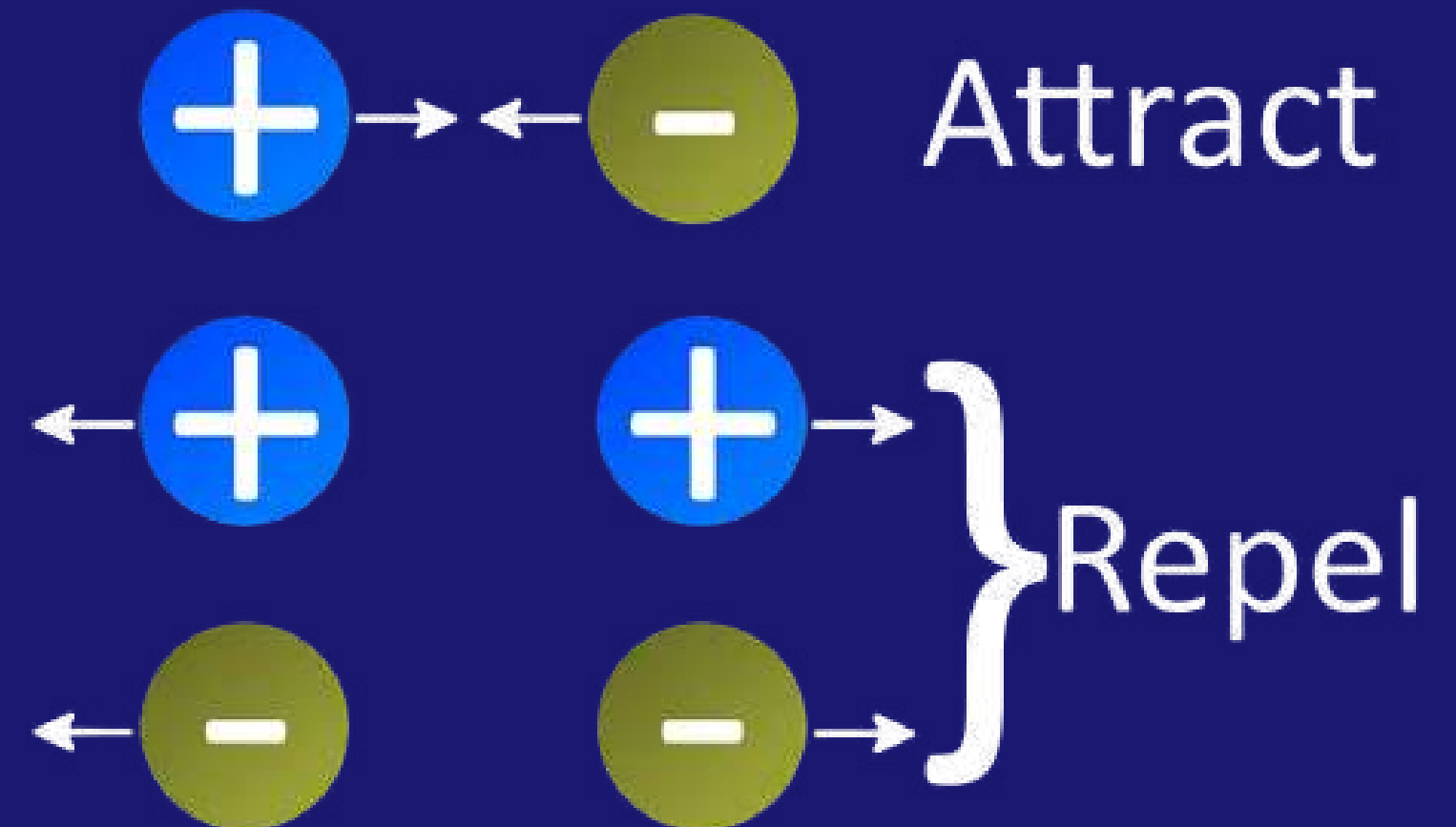
ELECTRIC CHARGE

Electric charge is a basic property of matter due to which a body can experience **electric force**.

Types of Electric Charge

1. **Positive charge**: Example: Proton has positive charge.
2. **Negative charge**: Example: Electron has negative charge.

Like charges repel each other and unlike charges attract each other.



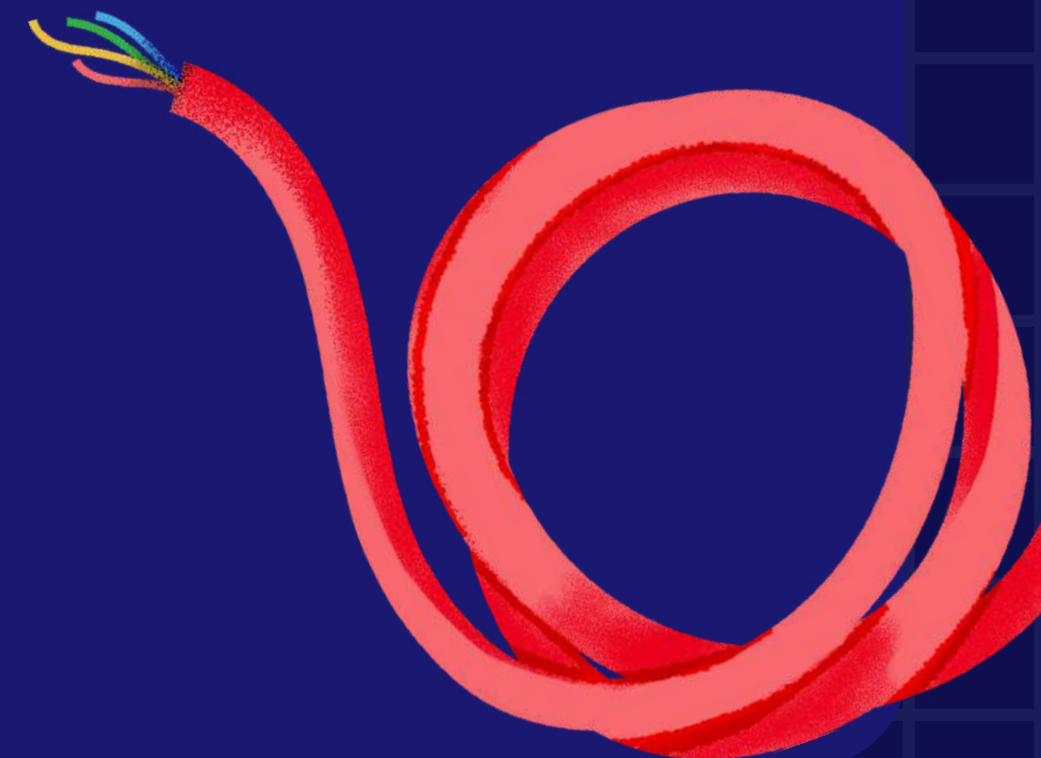
Charge in Metallic Wires

In metallic wires, electric charge flows due to the movement of electrons.
So, in a metal wire: **Flow of electrons = Flow of electric charge**

The SI unit of electric charge is coulomb, written as C.

1 coulomb charge = charge contained in nearly 6×10^{18} electrons.

Charge on one electron: $1.6 \times 10^{-19} \text{C}$



Basic Properties of Electric Charge

1. **Additivity of Charge:** The total charge on a body or system is the algebraic sum of all individual charges present on it.

Example: If a body has charges +2 C and -1 C, then total charge is: $+2C + (-1C) = +1C$

2. **Conservation of Charge:** Electric charge can neither be created nor destroyed. It can only be transferred from one body to another.

Example: When a body loses electrons, it becomes positively charged.

When a body gains electrons, it becomes negatively charged.

3. **Quantisation of Charge:** Any observable charge on a body is always an integral multiple of the charge of an electron.

$$Q = \pm ne$$



Where,

Q = total charge on the body

n = number of electrons transferred, where $n = 1, 2, 3, 4, \dots$

e = charge on one electron = $1.6 \times 10^{-19} \text{C}$

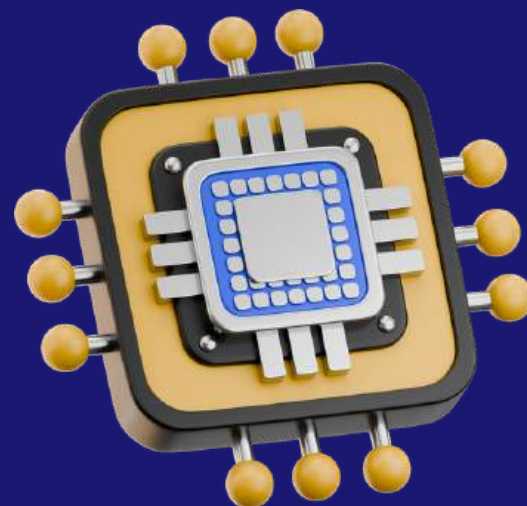
So, charge can have values like: $\pm 1e, \pm 2e, \pm 3e$ but not fractions like: $0.5e$ or $1.5e$

ELECTRICAL SUBSTANCES

Conductors	Semiconductors	Insulators
Allow electric current to flow easily.	Have electrical conductivity between conductors and insulators.	Do not allow electric current to flow easily.
Contain free electrons.	Conductivity can be altered by adding impurities or changing temperature.	Lack free electrons.
Examples: Copper, aluminum.	Examples: Silicon, germanium.	Examples: Rubber, plastic.



Gold



Wood

Q. A body gains 5×10^{12} electrons. Calculate the total charge acquired by the body.

Given: Charge on one electron = 1.6×10^{-19} C

Q. In a wire, 1.2×10^{20} electrons flow from one end to the other. Calculate the total charge transferred.

[NCERT]

Q. Calculate the number of electrons constituting one coulomb of charge

Q. Assertion (A): The unit of electric charge is coulomb.

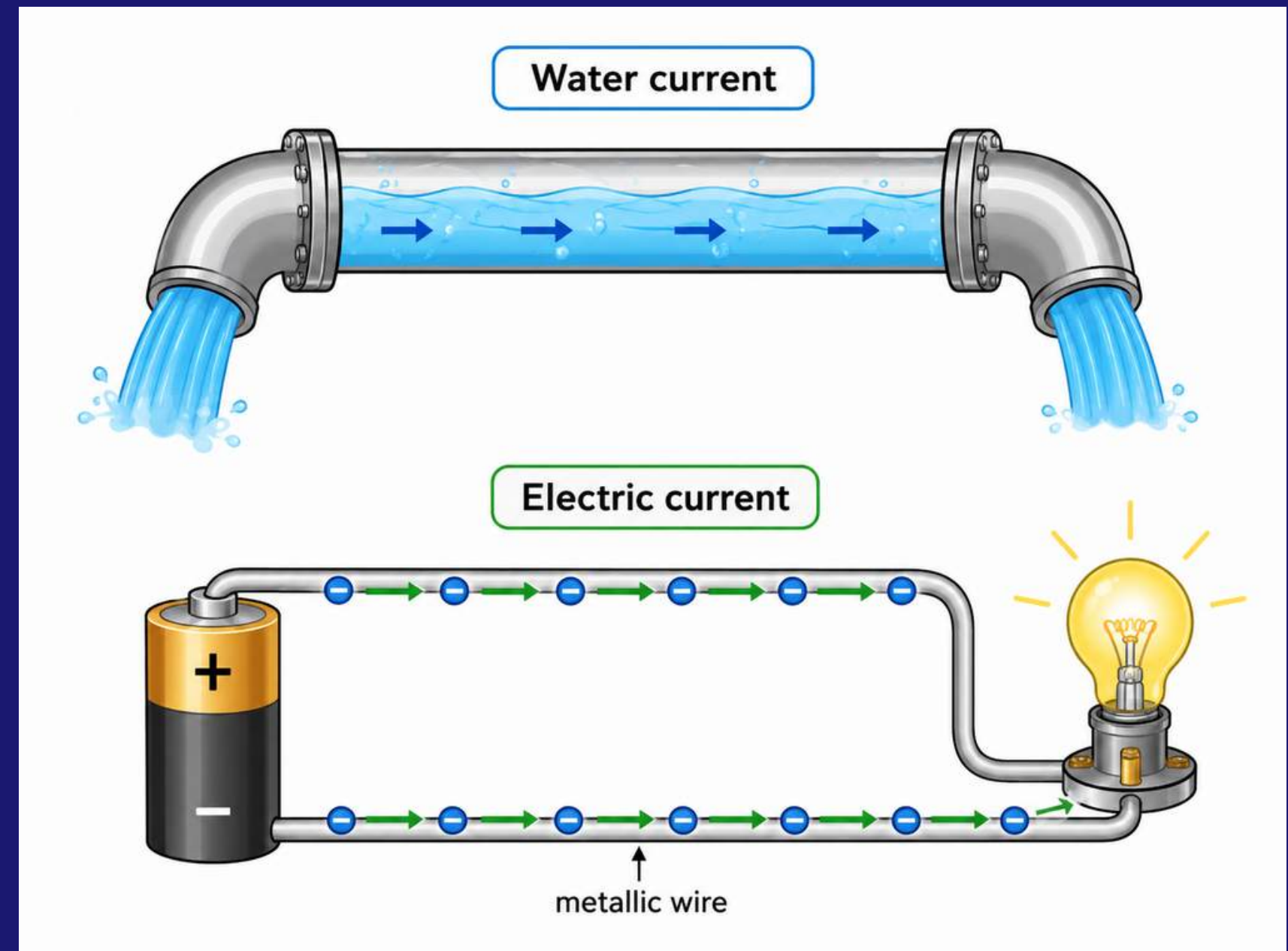
Reason (R): One coulomb is the charge carried by 6.25×10^{18} electrons.

**Q. If a body has a deficiency of electrons, what will be the nature of charge on it? Justify.
[CBSE 2016, 2022]**

ELECTRIC CURRENT

When electric charge flows through a conductor, such as a metallic wire, it is called **electric current**.

Example: In a torch, the cell or battery provides the flow of charges through the bulb, due to which the bulb glows.



Electric current is the amount of charge flowing through any cross-section of a conductor in unit time.

Or,

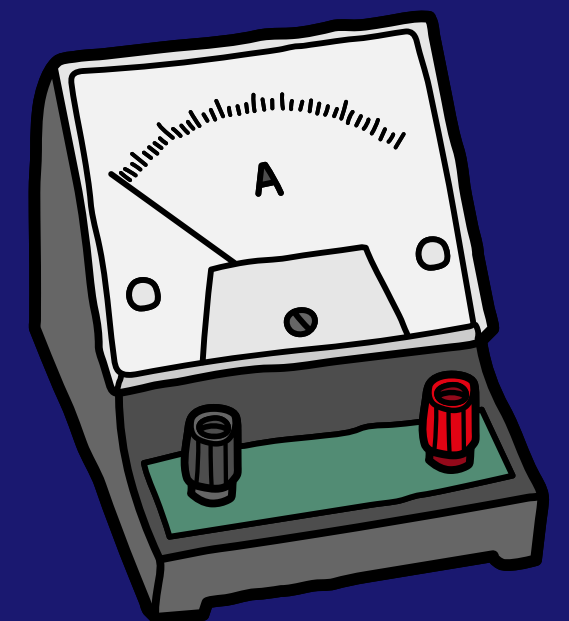
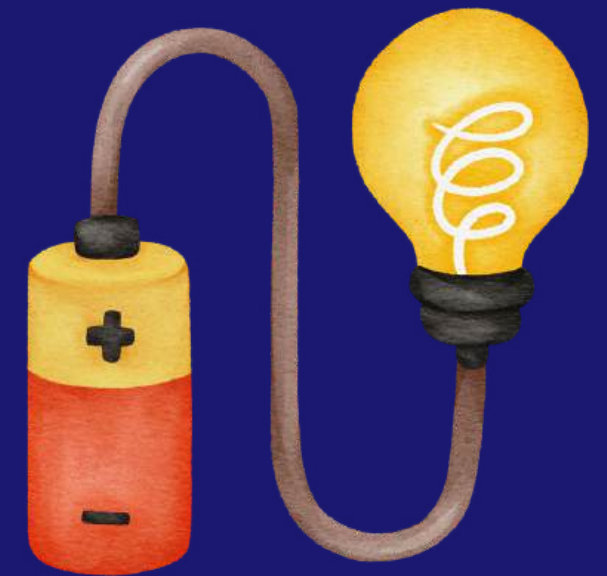
Electric current is the rate of flow of electric charges.

$$\text{Current} = \frac{\text{Amount of charge}}{\text{time}}$$

$$I = \frac{Q}{t}$$

The SI unit of electric current is **ampere**, written as A.

*Electric current is measured by an instrument called an **ammeter**.*



If 1 coulomb of charge flows through a conductor in 1 second, then the current is called **1 ampere**.

$$1\text{A} = 1\text{Cs}^{-1}$$

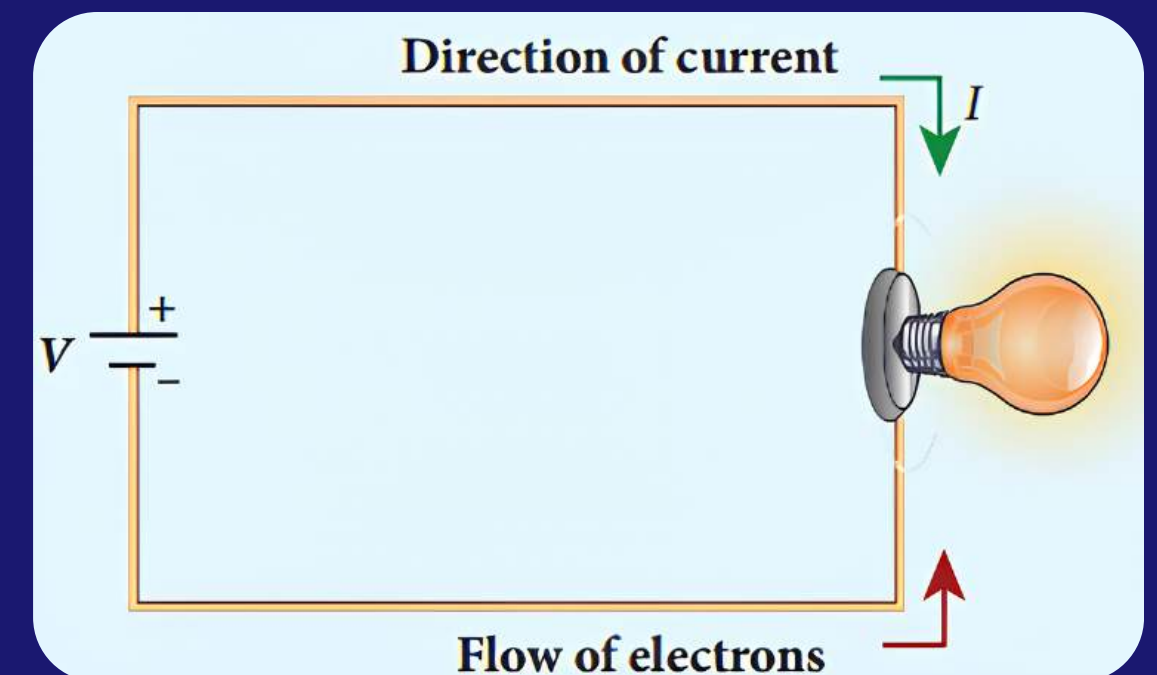
milliampere $\rightarrow 1\text{mA} = 10^{-3}\text{A}$

microampere $\rightarrow 1\mu\text{A} = 10^{-6}\text{A}$

In metals: Electric current is due to the flow of electrons.

Direction of electric current is opposite to the direction of flow of electrons.

In an electric circuit, electric current flows from the positive terminal of the cell to the negative terminal of the cell.

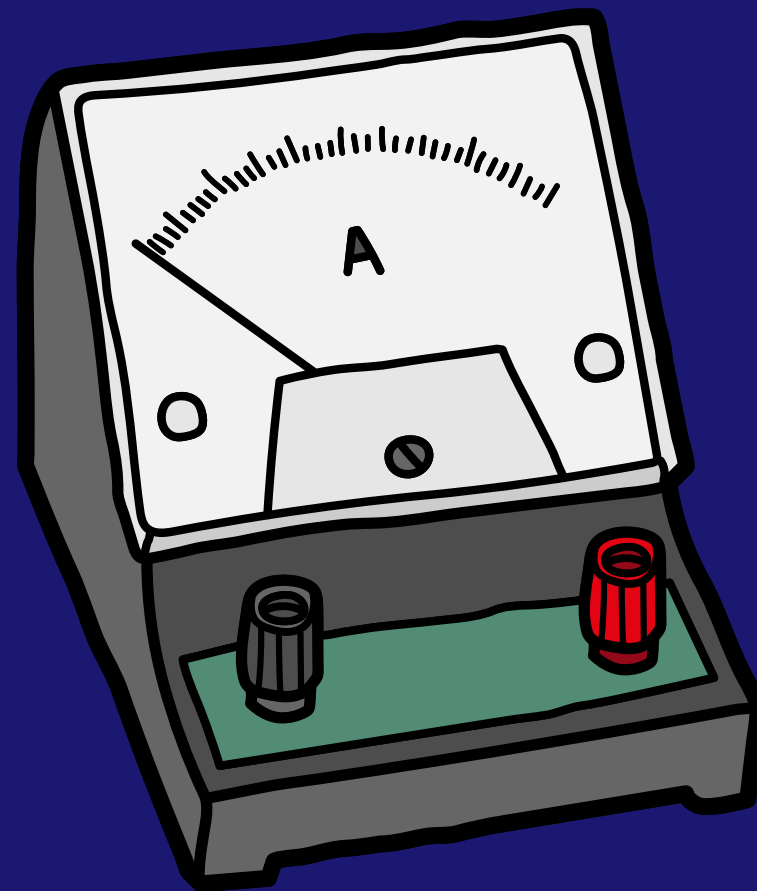


Ammeter:

An ammeter is an instrument used to measure electric current in a circuit.

Electric current is measured in ampere (A), so the reading of an ammeter is also in ampere.

An ammeter is always connected in **series** in the circuit.



[NCERT]

Q. A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit.

Q. If 1800 C of charge flows through a circuit in 30 minutes, calculate the current.
[CBSE 2018, 2024]

Q. A current of 5 A flows through a conductor for 3 minutes.

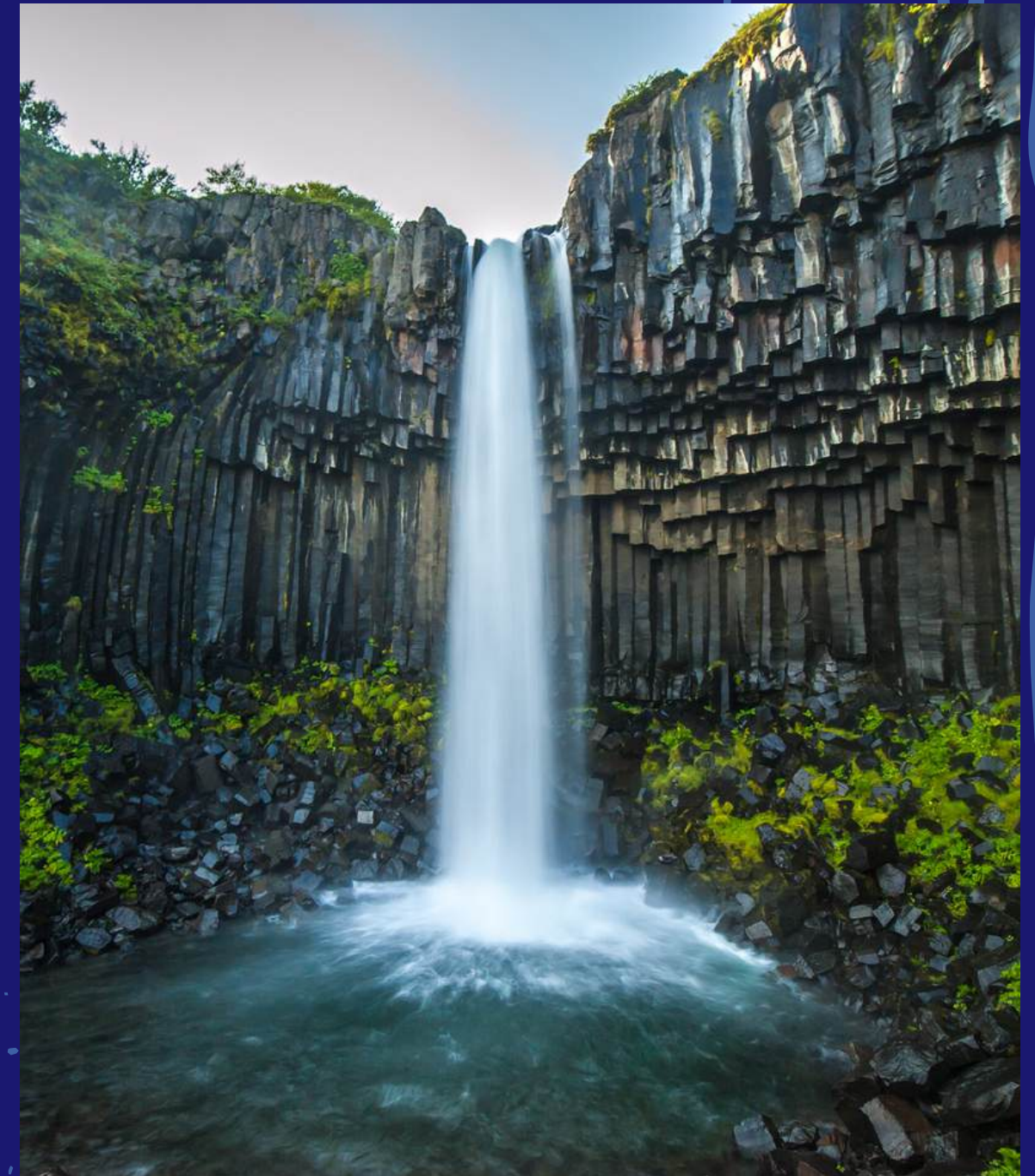
(i) Calculate the amount of charge passed through any area of cross-section of the conductor.

(ii) If the charge of an electron is 1.6×10^{-19} C, then calculate the total number of electrons flowing.

ELECTRIC POTENTIAL AND POTENTIAL DIFFERENCE

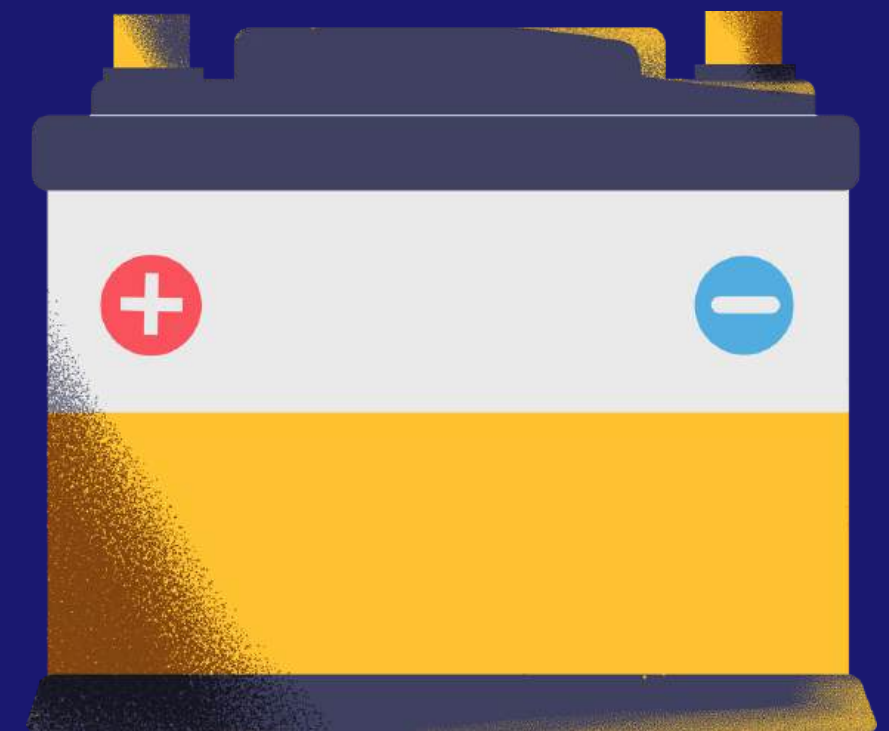
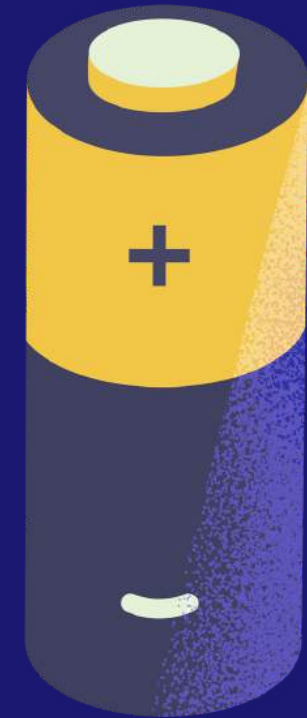
Why do charges flow?

- Electric charges do not flow in a conductor by themselves.
- To make charges flow through a metallic wire, a **potential difference** is required between the two ends of the conductor.
- Just like water flows due to pressure difference, electric charges flow due to potential difference.



Role of Cell or Battery

- A **cell or battery** produces potential difference across its terminals.
- This potential difference pushes the charges and makes them move through the conductor.
- Due to this movement of charges, electric current is produced.
- The cell uses its **stored chemical energy** to maintain the flow of current in the circuit.



POTENTIAL DIFFERENCE

Electric potential difference between two points is defined as the work done to move a unit charge from one point to another.

$$V = W/Q$$

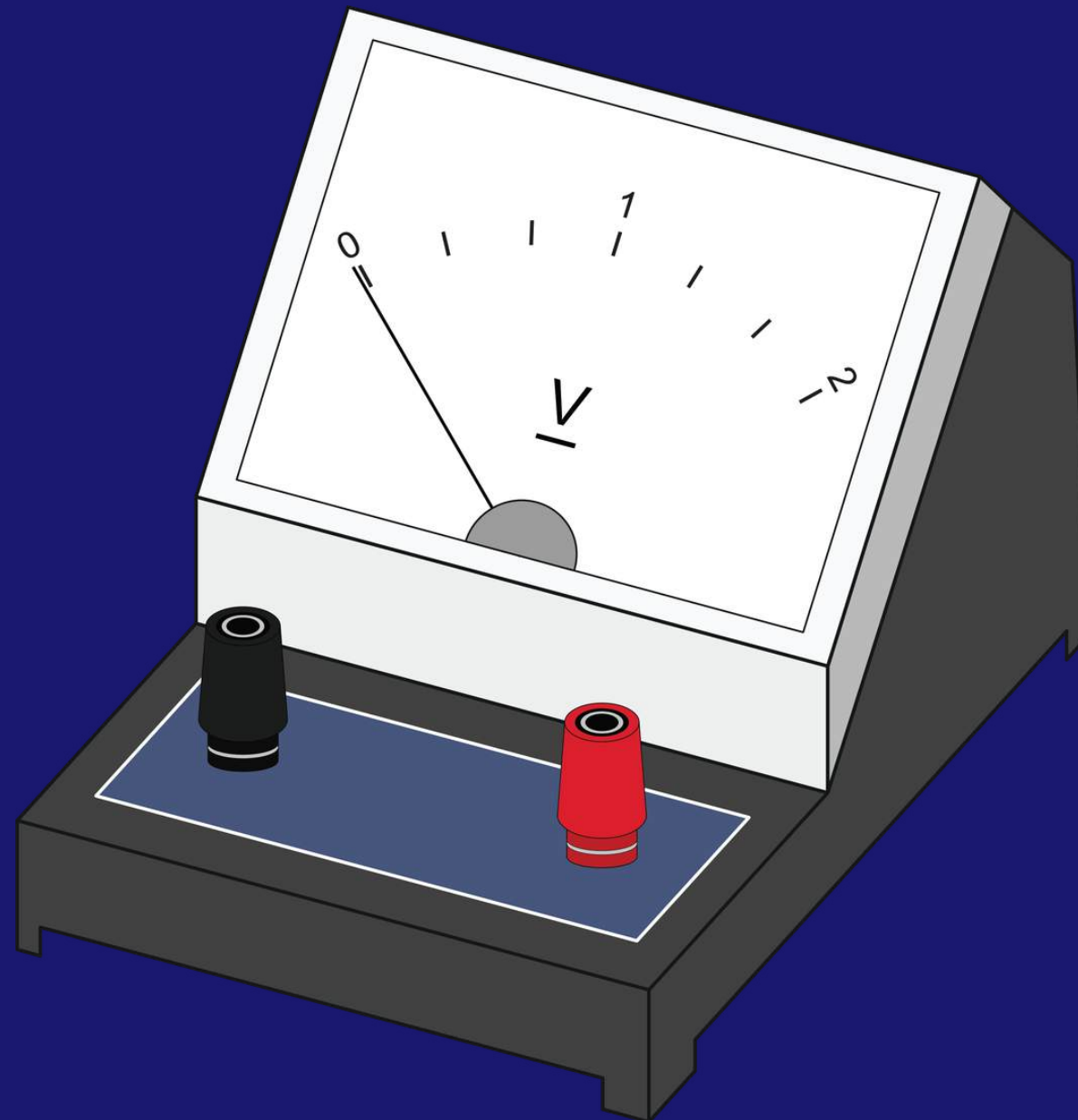
The SI unit of potential difference is **volt**, written as V.

Potential difference between two points is said to be **1 volt** if 1 joule of work is done to move 1 coulomb of charge from one point to another.

$$V = 1 \text{ J C}^{-1}$$

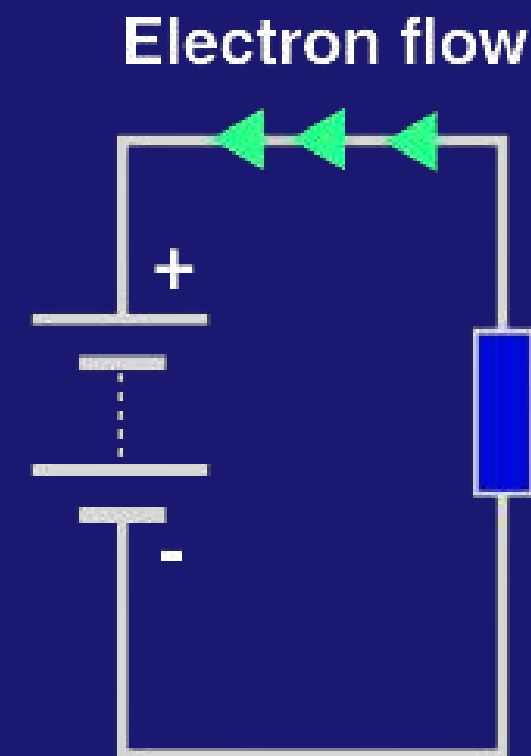
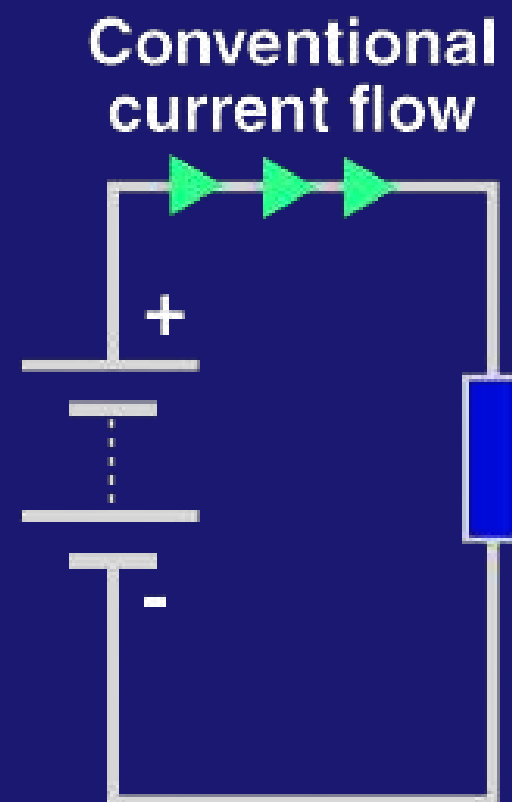
Voltmeter:

- Potential difference is measured by an instrument called a voltmeter.
- A voltmeter is always connected in **parallel** across the points between which potential difference is to be measured.



WHY CURRENT FLOWS

- **Electrons (e^-)** flow in a conductor from lower potential to higher potential.
- **Conventional current** flows from higher potential to lower potential.
- A **cell or battery** creates this potential difference (V), which makes charges flow in a closed circuit.



Q. The work done in moving a charge from one point to another is 20 J. If the potential difference between the points is 10 V, what is the charge?

[NCERT]

Q. How much work is done in moving a charge of 2 C across two points having a potential difference 12 V?

[NCERT]

Q. How much energy is given to each coulomb of charge passing through a 6 V battery?

Q. What is the potential difference if 30 J of work is done to move 5 C of charge?

- a) 5 V**
- b) 6 V**
- c) 150 V**
- d) 25 V**

- Q. (a) What is meant by saying that the potential difference is 10 V?**
- (b) How much work is done in moving a charge of 4 C across two points having 10 V potential difference?**
- (c) Name the device used to measure this quantity and state how it is connected in a circuit.**



***“A LITTLE
PROGRESS EACH
DAY ADDS UP TO
BIG RESULTS.”***